Statistical Mechanics Homework (Chapter2)

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题目 1. 2-2 证明以下各式:

(a) $\left(\dfrac{\partial U}{\partial p}\right)_V = -T\left(\dfrac{\partial V}{\partial T}\right)_S$;

(b) $\left(\dfrac{\partial U}{\partial V}\right)_p = T\left(\dfrac{\partial p}{\partial T}\right)_S - p$;

(c) $\left(\dfrac{\partial U}{\partial p}\right)_T = -T\left(\dfrac{\partial V}{\partial T}\right)_p - p\left(\dfrac{\partial V}{\partial p}\right)_T$;

(d) $U = -T^2\left(\dfrac{\partial (F/T)}{\partial T}\right)_V$;

(e) $H = -T^2\left(\dfrac{\partial (G/T)}{\partial T}\right)_p$;

(f) $\left(\dfrac{\partial T}{\partial V}\right)_S = -\dfrac{T}{C_V}\left(\dfrac{\partial p}{\partial T}\right)_V$;

(g) $\left(\dfrac{\partial T}{\partial p}\right)_S = \dfrac{TV\alpha}{C_p}$.

在 (g) 中, α 为定压膨胀系数. 由此证明水在 0 °C 与 4 °C 之间有 $\left(\dfrac{\partial T}{\partial p}\right)_S < 0$, 这说明什么问题?

题目 1 的解答. (a)

$$\begin{aligned}\text{RHS} &= -\left(\frac{\partial U}{\partial S}\right)_V\left(\frac{\partial V}{\partial T}\right)_S \\ &= \left(\frac{\partial U}{\partial S}\right)_V\left(\frac{\partial S}{\partial p}\right)_V = \left(\frac{\partial U}{\partial p}\right)_V = \text{LHS}\end{aligned}$$

(b)

$$\begin{aligned}\text{RHS} &= \left(\frac{\partial U}{\partial S}\right)_V\left(\frac{\partial S}{\partial V}\right)_p + \left(\frac{\partial U}{\partial V}\right)_S \\ &= \left(\frac{\partial U}{\partial V}\right)_p = \text{LHS}\end{aligned}$$

(c)

$$\begin{aligned}\text{RHS} &= -\left(\frac{\partial U}{\partial S}\right)_p \left(\frac{\partial V}{\partial T}\right)_p + \left(\frac{\partial U}{\partial V}\right)_S \left(\frac{\partial V}{\partial p}\right)_T \\ &= \left(\frac{\partial U}{\partial S}\right)_p \left(\frac{\partial S}{\partial p}\right)_T + \left(\frac{\partial U}{\partial V}\right)_S \left(\frac{\partial V}{\partial p}\right)_T \\ &= \left(\frac{\partial U}{\partial p}\right)_T = \text{LHS}\end{aligned}$$

(d)

$$\begin{aligned}\text{RHS} &= -T^2 \left(\left[\frac{\partial (U/T)}{\partial T}\right]_V - \left(\frac{\partial S}{\partial T}\right)_V\right) \\ &= -T^2 \left(-\frac{U}{T^2} + \frac{1}{T}\left(\frac{\partial U}{\partial T}\right)_V - \left(\frac{\partial S}{\partial T}\right)_V\right) \\ &= U + T^2 \left(\left(\frac{\partial S}{\partial U}\right)_V \left(\frac{\partial U}{\partial T}\right)_V - \left(\frac{\partial S}{\partial T}\right)_V\right) \\ &= U = \text{LHS}\end{aligned}$$

(e)

$$\begin{aligned}\text{RHS} &= -T^2 \left(\left[\frac{\partial (H/T)}{\partial T}\right]_p - \left(\frac{\partial S}{\partial T}\right)_p\right) \\ &= -T^2 \left(-\frac{H}{T^2} + \frac{1}{T}\left(\frac{\partial H}{\partial T}\right)_p - \left(\frac{\partial S}{\partial T}\right)_p\right) \\ &= H + T^2 \left(\left(\frac{\partial S}{\partial H}\right)_p \left(\frac{\partial H}{\partial T}\right)_p - \left(\frac{\partial S}{\partial T}\right)_p\right) \\ &= H = \text{LHS}\end{aligned}$$

(f)

$$\begin{aligned}\text{RHS} &= -\frac{\left(\frac{\partial U}{\partial S}\right)_V}{\left(\frac{\partial U}{\partial T}\right)_V}\left(\frac{\partial p}{\partial T}\right)_V \\ &= -\left(\frac{\partial U}{\partial S}\right)_V \left(\frac{\partial p}{\partial U}\right)_V \\ &= -\left(\frac{\partial p}{\partial S}\right)_V = \left(\frac{\partial T}{\partial V}\right)_S = \text{LHS}\end{aligned}$$

(g)

$$\begin{aligned}\text{RHS} &= \frac{\left(\frac{\partial H}{\partial S}\right)_p V \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p}{\left(\frac{\partial H}{\partial T}\right)_p} \\ &= \left(\frac{\partial V}{\partial S}\right)_p = \left(\frac{\partial T}{\partial p}\right)_S = \text{LHS}\end{aligned}$$

水在 0 °C 与 4 °C 之间, 水具有异常热胀冷缩行为——随温度升高密度先增加而后减, 因此此时 $\alpha < 0$. 同时由于 $C_p, T, V > 0$, 则 $\left(\dfrac{\partial T}{\partial p}\right)_S < 0$.

这说明在这个区间内做等熵过程, 加压的同时温度还会降低. 这是水的特殊结构导致的.

题目 2. 2-3 证明每物质的量范德瓦尔斯气体, 有

$$c_p - c_V = \frac{R}{1 - \dfrac{2a(v-b)^2}{RTv^3}}$$

题目 2 的解答. 由定义, 不难有

$$c_p = \left(\frac{\partial H}{\partial T}\right)_p, c_V = \left(\frac{\partial U}{\partial T}\right)_V$$

因此我们想到

$$\begin{aligned}
\mathrm{d}U &= T\mathrm{d}S - p\mathrm{d}V \\
&= T\left[\left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial S}{\partial V}\right)_T \mathrm{d}V\right] - p\mathrm{d}V \\
&= T\left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left[T\left(\frac{\partial S}{\partial V}\right)_T - p\right]\mathrm{d}V
\end{aligned}$$

因此, 有

$$c_V = T\left(\frac{\partial S}{\partial T}\right)_V$$

同理有

$$c_p = T\left(\frac{\partial S}{\partial T}\right)_p$$

由此, 有

$$\begin{aligned}
c_p - c_V &= T\left[\left(\frac{\partial S}{\partial T}\right)_p - \left(\frac{\partial S}{\partial T}\right)_V\right] \\
&= T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \\
&= T\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_p
\end{aligned}$$

现在, 我们再代入物态方程

$$\left(p + \frac{a}{v^2}\right)(v-b) = RT$$

化简后, 不难得证

$$c_p - c_V = \frac{R}{1 - \dfrac{2a(v-b)^2}{RTv^3}}$$

题目 3. 2-4 证明:

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T\left(\frac{\partial^2 p}{\partial T^2}\right)_V$$

$$
\begin{aligned}
\mathrm{LHS} &= \left(\frac{\partial\left[T\left(\frac{\partial S}{\partial T}\right)_V\right]}{\partial V}\right)_T \\
&= T\left(\frac{\partial\left[\left(\frac{\partial S}{\partial T}\right)_V\right]}{\partial V}\right)_T \\
&= T\left(\frac{\partial\left[\left(\frac{\partial S}{\partial V}\right)_T\right]}{\partial T}\right)_V \\
&= T\left(\frac{\partial\left[\left(\frac{\partial p}{\partial T}\right)_V\right]}{\partial T}\right)_V \\
&= T\left(\frac{\partial^2 p}{\partial T^2}\right)_V = \mathrm{RHS}
\end{aligned}
$$

题目 4. 2-5 计算 1mol 范德瓦尔斯气体的内能和熵作为温度和体积的函数.

题目 4 的解答.

$$
\begin{aligned}
\mathrm{d}S &= \left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial S}{\partial v}\right)_T \mathrm{d}v \\
&= \frac{c_V}{T}\mathrm{d}T + \left(\frac{\partial p}{\partial T}\right)_v \mathrm{d}v \\
&= \frac{c_V}{T}\mathrm{d}T + \frac{R}{v-b}\mathrm{d}v
\end{aligned}
$$

两边积分, 有

$$
S = \int \frac{c_V}{T}\mathrm{d}T + R\ln(v-b) + S_0
$$

$$
\begin{aligned}
\mathrm{d}U &= T\mathrm{d}S - p\mathrm{d}v \\
&= T\left(\frac{c_V}{T}\mathrm{d}T + \frac{R}{v-b}\mathrm{d}v\right) - p\mathrm{d}v \\
&= c_V\mathrm{d}T - \left(p - \frac{RT}{v-b}\right)\mathrm{d}v \\
&= c_V\mathrm{d}T + \left(\frac{a}{v^2}\right)\mathrm{d}v
\end{aligned}
$$

两边积分, 有

$$
U = \int c_V\mathrm{d}T - \frac{a}{v} + U_0
$$

题目 5. 2-6 对某种气体的膨胀系数和压缩系数进行测量的结果如下:

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{R}{p} + \frac{a}{T^2}, \left(\frac{\partial V}{\partial p}\right)_T = -Tf(p)$$

其中,a 为常量,$f(p)$ 只是 p 的函数. 在低压下,1mol 气体的定压热容为 $\frac{5}{2}R$. 证明:

(a)$f(p) = \frac{R}{p^2}$;(b) 物态方程为 $pV = RT - \frac{ap}{T}$;(c)$C_p = \frac{2ap}{T^2} + \frac{5}{2}R$.

题目 5 的解答. (a) 由偏导算子可对易, 有

$$\left(\frac{\partial\left[\frac{R}{p} + \frac{a}{T^2}\right]}{\partial p}\right)_T = \left(\frac{\partial\left(-Tf(p)\right)}{\partial T}\right)_p$$

化简即可得

$$f(p) = \frac{R}{p^2}$$

(b)

$$\begin{aligned}
\mathrm{d}V &= \left(\frac{\partial V}{\partial T}\right)_p \mathrm{d}T + \left(\frac{\partial V}{\partial p}\right)_T \mathrm{d}p \\
&= \left(\frac{R}{p} + \frac{a}{T^2}\right)\mathrm{d}T - \frac{RT}{p^2}\mathrm{d}p \\
&= \mathrm{d}\left(\frac{RT}{p} - \frac{a}{T}\right)
\end{aligned}$$

因此,$V = \frac{RT}{p} - \frac{a}{T} + C$. 当 $T \to \infty$, 由于退化为理想气体, 则 $C = 0$.

综上, 证明到, 物态方程为 $pV = RT - \frac{ap}{T}$.

(c)

$$\begin{aligned}
\mathrm{d}S &= \left(\frac{\partial S}{\partial T}\right)_p \mathrm{d}T + \left(\frac{\partial S}{\partial p}\right)_T \mathrm{d}p \\
&= \frac{C_p}{T}\mathrm{d}T - \left(\frac{\partial V}{\partial T}\right)_p \mathrm{d}p \\
&= \frac{C_p}{T}\mathrm{d}T - \left(\frac{R}{p} + \frac{a}{T^2}\right)\mathrm{d}p
\end{aligned}$$

由于偏导算子可对易, 则

$$\left(\frac{\partial(C_p/T)}{\partial p}\right)_T = \left(\frac{\partial}{\partial T}\left(\frac{R}{p} + \frac{a}{T^2}\right)\right)_p$$

化简后, 即有

$$\left(\frac{\partial C_p}{\partial p}\right)_T = \frac{2a}{T^2}$$

因此, 有

$$\mathrm{d}C_p = \frac{2a}{T^2}\mathrm{d}p + \varphi(p,T)\mathrm{d}T$$

由偏导算子可对易, 有

$$-\frac{4a}{T^3}=\left(\frac{\partial\varphi}{\partial p}\right)_T$$

两侧积分后, 有

$$\varphi(p,T)=-\frac{4ap}{T^3}+\phi(T)$$

代入, 整理有

$$\mathrm{d}C_p=\mathrm{d}\left(\frac{2ap}{T^2}+\Psi(T)\right)$$

因此,

$$C_p=\frac{2ap}{T^2}+\Psi(T)$$

当 $p\to 0$ 时,$C_p\to\Psi(T)=\dfrac{5}{2}R$. 综上, 有

$$C_p=\frac{2ap}{T^2}+\frac{5}{2}R$$